

Customer Behavior Analysis Report

This project focuses on analyzing customer transaction and behavioral data to identify customer segments, purchase patterns, and churn risks. The analysis was performed using Python, Pandas, Matplotlib, Seaborn, and Machine Learning techniques such as K-Means clustering. The goal of the project is to help businesses understand customer behavior and improve customer engagement strategies.

1. Data Loading and Cleaning

The dataset used in this project contains approximately 1000 customer transaction records including customer IDs, purchase amounts, transaction dates, and product details. During the data cleaning stage, missing values were handled using fillna(), duplicate rows were removed, and incorrect data types were converted into appropriate formats such as datetime. This stage ensured that the dataset was accurate, consistent, and ready for analysis.

2. Feature Engineering

Feature engineering was performed to create meaningful business metrics from the raw dataset. Features such as total purchase value, customer frequency, recency score, loyalty score, and estimated churn risk were generated. These engineered features helped improve the quality of customer segmentation and provided deeper insights into customer purchasing behavior.

3. Customer Segmentation using K-Means Clustering

K-Means clustering was used to group customers into different behavioral segments based on spending patterns and engagement levels. The customers were divided into clusters such as high-value customers, loyal customers, occasional buyers, and high churn-risk customers. This segmentation helps businesses create personalized marketing strategies for different customer groups.

Customer Segment	Characteristics	Business Strategy
High-Value Customers	Frequent and high spending	Provide premium offers
Loyal Customers	Consistent purchases	Reward loyalty points
Occasional Buyers	Rare purchases	Send promotional campaigns
Churn Risk Customers	Low engagement	Offer retention discounts

4. Visualizations and Insights

Several visualizations were created to analyze customer behavior trends. Charts such as sales distribution, monthly purchase trends, customer frequency distribution, and cluster visualizations were generated using Matplotlib and Seaborn. The visual analysis revealed that a small group of customers contributed significantly to the total revenue, while some customers showed declining engagement over time.

5. Recommendations

- Provide personalized offers for high-value customers to improve retention.
- Launch loyalty programs to increase repeat purchases.
- Use targeted marketing campaigns for occasional buyers.

- Monitor churn-risk customers regularly and offer retention discounts.
- Use automated dashboards for continuous monitoring of customer behavior.

Conclusion

The customer behavior analysis project successfully identified important customer segments and purchase patterns using data analytics and machine learning techniques. The findings can help businesses make data-driven decisions, improve customer engagement, and increase revenue through targeted strategies. This project also demonstrates practical skills in data cleaning, feature engineering, visualization, clustering, and business insight generation.